# RepIt – Firmware Feature Specification

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## 1. System Overview

RepIt consists of **two identical wearable devices** that:

- Capture synchronized motion data
- Communicate with each other wirelessly
- Independently connect to a mobile app via BLE
- Stream structured motion frames in real time
- Operate on low power for full-day usability

## 2. Hardware-Level Firmware Control

### 2.1 Microcontroller / SoC

**Requirements:**

- BLE 5.x compliant SoC (final selection per engineer recommendation)
- Must support:
  - Multiple low power modes (sleep, deep sleep, system off)
  - External interrupt handling (IMU + touch interface)
  - OTA firmware update capability over BLE
  - Stable BLE streaming at minimum 50 Hz (target scalable to 100 Hz)
  - Capacitive touch sensing support (integrated or via external IC)
  - Sufficient flash and RAM for dual-device sync + OTA + buffering

**Performance Targets:**

- Stable data throughput with no packet loss under moderate athletic motion
- OTA-safe memory partitioning (bootloader + application separation)

### 2.2 IMU (6-Axis)

**Sensor Type:**

- 3-axis accelerometer
- 3-axis gyroscope

**Configurable Parameters (via firmware):**

- Sampling rate: 25 Hz / 50 Hz / 100 Hz
- Measurement range: ±2g, ±4g, ±8g, ±16g
- Gyro range configurable if supported

**Firmware Requirements:**

- Interrupt-driven motion detection
- FIFO buffering utilization (if supported by IMU)
- Timestamp association per sample (generated by MCU timer)
- Basic calibration routine:
    - Offset calibration
    - Static bias compensation
- Drift mitigation strategy (firmware-level filtering or correction readiness)

### 2.2.1 Sensor Calibration & Offset Management

**Firmware Requirements:**

**Factory Calibration (If feasible in V1)**

- Store static offset values in non-volatile memory
- Support calibration routine triggered via BLE command

**Runtime Offset Handling**

- Apply accelerometer offset correction before transmission
- Apply gyroscope bias correction
- Support zero-motion recalibration routine
- Store calibration coefficients in flash memory

**Drift Handling**

- Basic drift mitigation logic (firmware filtering readiness)
- Provide raw + corrected values (configurable mode)

**Calibration Trigger Options**

- Via mobile app command
- Via touch gesture (long touch + hold sequence)
- Automatic calibration during first 5 seconds of idle state

## 2.3 Power Management

### Required Firmware Behaviors

- Deep sleep mode (target: low μA current consumption)
- IMU interrupt-based wake
- Capacitive touch interrupt-based wake
- Automatic sleep after configurable inactivity timeout
- Battery monitoring:
    - Periodic voltage sampling via ADC
    - Percentage estimation algorithm
    - Battery percentage exposed via BLE
- Low battery threshold alert (BLE + LED indication)
- Critical battery threshold:
    - Graceful shutdown sequence
    - Data flush before shutdown

## 2.4 Charging System (Magnetic Pogo – Dual Simultaneous Charging)

### Hardware Context:

- Devices magnetically attach together for charging
- Pogo connector interface
- Both units charge simultaneously when docked

### Firmware Requirements:

- Charge detection via pogo connector sensing
- Charging state reporting via BLE:
    - Charging
    - Fully charged
    - Not charging
- Magnetic attach detection (if electrically detectable)
- Automatic transition to low-power charging mode
- Prevent session start while charging (configurable behavior)

## 2.5 LED State Logic (Multi-Color LED)

Firmware must manage multi-color LED (RGB or dual-color depending on hardware).

**Required LED States:**

Pairing mode (BLE)

Charging status (fully charge/ running low)

Pairing status (Paired/ disconnected)

### 2.6 Capacitive Touch Interface (Replacing Physical Button)

The device will use a capacitive touch surface instead of a mechanical button.

**Firmware Requirements:**

- Touch interrupt-driven detection
- Debounce handling
- False-trigger rejection (sweat / moisture mitigation)
- Support for gesture differentiation:

**Touch Gesture Mapping (Proposed – Adjustable)**

- Single Tap → Start Session
- Double Tap → Stop Session
- Long Touch → Pairing mode

Firmware must:

- Confirm gesture detection via LED feedback
- Prevent accidental session start during intense movement
- Include configurable gesture timing thresholds

## 3. Dual-Device Synchronization Firmware

This is a core system requirement.

### 3.1 Pairing Between Devices

- Secure initial device-to-device pairing

- Store bonded partner ID in flash
- Automatic reconnection when powered on

### 3.2 Role Architecture

Devices must support:

- Dynamic role assignment:
  - Temporary Sync Leader
  - Sync Peer
- Automatic failover if leader disconnects
- No permanent master

### 3.3 Time Synchronization

- Timestamp alignment protocol between devices
- Periodic re-sync (configurable interval)
- Drift compensation handling
- Maximum allowable timestamp drift threshold defined

Target:

- < 5 ms inter-device timestamp deviation

### 3.4 Sync Trigger

- Event-based sync handshake before session start
- Sync validation confirmation
- Session start command broadcast

## 4. BLE Communication With Mobile App

Each device must:

- Maintain independent BLE connection to app
- Expose:
  - Device ID
  - Firmware version
  - Battery level
  - Session state
  - IMU data stream
  - Error states

### 4.1 BLE Services (Custom GATT)

Custom RepIt Service:

Characteristics:

1. Control Characteristic (Write)
    - Start session
    - Stop session
    - Request calibration
    - Sync devices
    - Set sampling rate
2. Data Characteristic (Notify)
    - Structured motion frames
    - Timestamp included
    - Sequence number included
3. Status Characteristic (Read/Notify)
    - Battery %
    - Charging state
    - Connection state
    - Error flags

### 4.2 Streaming Requirements

- Minimum 50Hz stable streaming
- No data loss under moderate athletic activity
- Configurable connection interval
- Packet size optimized for throughput
- Re-transmission or sequence validation mechanism

## 5. Motion Data Structure (Critical)

Each motion frame must include:

- Device ID
- Timestamp (ms precision)
- Sequence Number
- Accel X/Y/Z
- Gyro X/Y/Z
- Battery %
- Flags (optional)

## 6. Session Logic (Application Layer)

Firmware must manage:

States:

- Idle
- Advertising
- Connected
- Synced
- Recording
- Streaming
- Sleeping
- Charging

Session behavior:

1. App sends “Start”
2. Devices sync
3. Recording begins
4. Windowed transmission
5. App sends “Stop”
6. Devices save final state
7. Return to idle

## 7. Error Handling & Reliability

Must include:

- IMU failure detection
- BLE disconnect recovery
- Partner device disconnect handling
- Battery critical shutdown logic
- Watchdog timer reset system
- Safe reboot recovery

Error states must be exposed via BLE.

## 8. Data Processing & ML Architecture

### 8.1 V1 Architecture (Mobile-Based ML)

- Devices stream raw motion data only
- No inference performed on-device
- ML model runs in mobile app

- Firmware must:
    - Maintain consistent sampling rate
    - Ensure synchronized timestamps between devices
    - Provide window-ready structured frames

Target window size:

- 1–3 second sliding window (configurable from app)

### 8.2 Data Requirements for ML

Firmware must ensure:

- Stable 50–100 Hz sampling
- Minimal packet loss
- Sequence numbering
- Timestamp precision <5 ms drift between devices
- No artificial smoothing unless enabled

### 8.3 Future: On-Device TinyML (Not Required in V1)

Architecture must remain compatible with:

- TensorFlow Lite Micro or equivalent
- Local inference capability
- Model storage in flash
- BLE command to switch between:
    - Raw streaming mode
    - Inference-only mode
    - Hybrid mode

→ Not required for first build, but memory planning must account for it.

## 9. Power Targets

V1 Targets:

- Minimum 8–12 hours active use
- Deep sleep < 100 μA target
- Optimized duty cycling

Firmware must:

- Minimize CPU wake time
- Use interrupt-driven logic
- Avoid polling loops

## 10. OTA Firmware Update (If Timeline Allows)

- Secure OTA via BLE
- Version validation
- Rollback protection
- Partial firmware update support (if feasible)